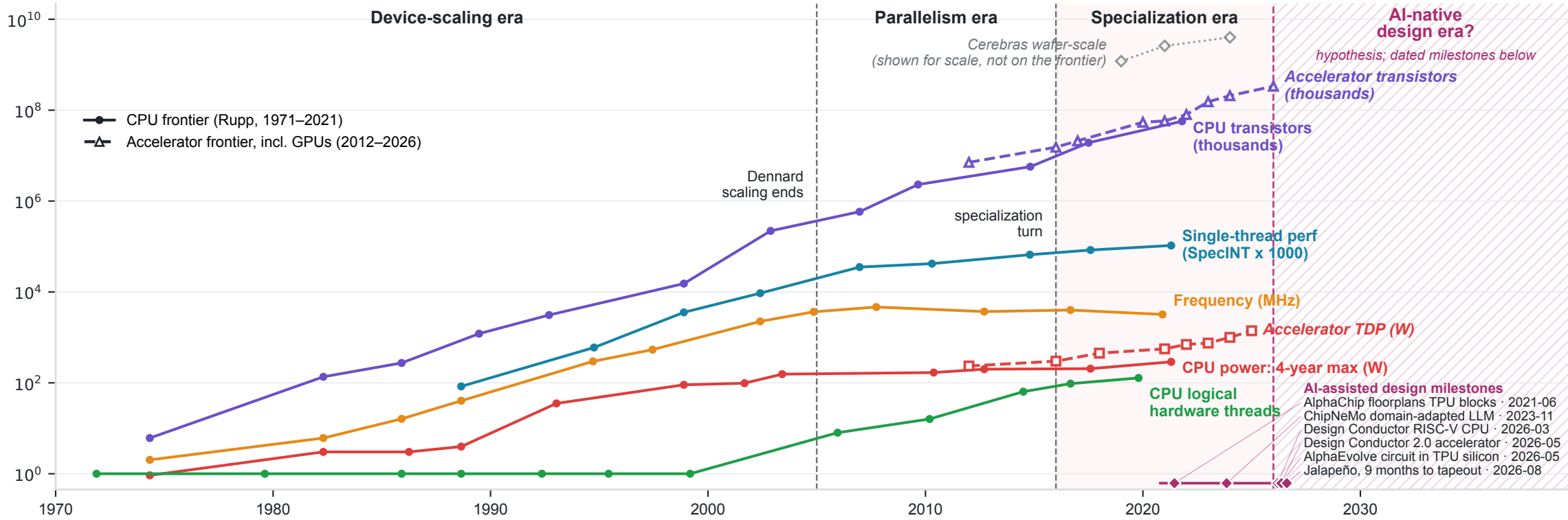

Device-scaling era
Parallelism era
Specialization era
AI-native design era?
hypothesis; dated milestones below
Cerebras wafer-scale (shown for scale, not on the frontier)
CPU frontier (Rupp, 1971–2021)
Accelerator frontier, incl. GPUs (2012–2026)
Dennard scaling ends
specialization turn
Accelerator transistors (thousands)
CPU transistors (thousands)
Single-thread perf (SpecINT x 1000)
Frequency (MHz)
Accelerator TDP (W)
CPU power: 4-year max (W)
CPU logical hardware threads
AI-assisted design milestones
AlphaChip floorplans TPU blocks · 2021-06
ChipNeMo domain-adapted LLM · 2023-11
Design Conductor RISC-V CPU · 2026-03
Design Conductor 2.0 accelerator · 2026-05
AlphaEvolve circuit in TPU silicon · 2026-05
Jalapeño, 9 months to tapeout · 2026-08
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
1970
1980
1990
2000
2010
2020
2030